Chair of Transportation Systems Engineering & Chair of Traffic Engineering
TUM School of Engineering and Design
Technical University of Munich

Microscopic Traffic Simulation — Calibration and Dynamic Applications

Winter Term 2025/26

Project Description

Topics	:	**Demand Calibration, Bus Stop-Skipping and Prioritization**
Due Date	:	**13/03/2026, 23:59**
Grading	:	**100 % of your final grade (80 % report, 20 % presentation)**

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1 Introduction

This document presents the graded project for the Microscopic Traffic Simulation — Calibration and Dynamic Applications module.

For the calibration project, you should work **individually**. You should complete and submit your own report and give an individual presentation. For the dynamic applications, you can work on the project in **groups of 2** (3 in case of exceptions) on a conceptual level. However, the actual implementation, flowcharts, result visualization and report writing should be finished and submitted **independently** with your own approach and words. You also must present your work **individually**. The weighting between the written report and the presentation is 80/20. A presentation is mandatory, i.e., we will not grade any report without presentation! Model calibration and dynamic applications will both contribute 50 % of the final grade.

The project should show the methods and tools you have developed in this module. There will be several consultation sessions in both demand calibration and dynamic applications. Please reserve your questions in the respective time slots, i.e. questions on Calibration on Tuesdays, questions on Dynamic Applications on Thursdays.

2 Important Guidelines for AI Use

2.1 Use of AI Tools (ChatGPT, Copilot, Gemini, etc.)

You are allowed to use AI tools (e.g., ChatGPT, Copilot, Gemini) **only for assistance**, such as:

- Language polishing/rewriting
- Coding

However, **full transparency is mandatory**. You must explicitly state:

- Which AI tool you used (e.g., “ChatGPT”)
- For what purpose (e.g., language proofreading, code suggestions)
- Which part of the output was influenced by AI (e.g., file name, function name, and line numbers)

Recommended format is to add an AI Acknowledgment section at the end of each part (e.g., Calibration / Dynamic Application). Example:

AI Acknowledgement - Calibration Module

I used ChatGPT (GPT-5) for language proofreading and to generate the initial version of the *estimate_parameters()* function (lines 80–110). I reviewed, modified, and verified the content myself.

2.2 Accountability and Academic Integrity

You are fully responsible for the technical correctness and originality of the report and supplementary files (code, results, figures). This includes:

- No fake or non-existent citations
- No placeholder text such as “below is the paragraph generated for your project”

- No unverified or hallucinated statements from AI

If we detect unacknowledged AI usage, fake citations, or AI-generated placeholder content, your project will fail.

3 Project Description

In the first part of the project, you will calibrate a given network based on available data (3.1). In the second part, you will create dynamic applications, namely implementing dynamic bus requests and the prioritization of buses in a signal controller (3.2).

You will find some input and script files for all these parts in the *Special Topics on Model Calibration* Moodle course. Please download the *MTS_25_26_model_input.zip* file and unpack it in your working directory.

The following subsections will describe the expected tasks in more detail.

3.1 Model Calibration (50 points in total)

This task should cover the calibration of a synthetic transportation model. Due to its large dimensions, the problem of calibrating the network is non-linear and complex. This task will help in understanding the involved complexity in calibrating a medium-sized traffic model and the importance of defining appropriate objective functions, SPSA parameters, and weights of different traffic data sources.

You are handed the following input data:

- A synthetic network to be calibrated, including the network file (*Aachen.net.xml*) and the TAZ file (*Aachen_taz.add.xml*).
- Prior OD matrices (*calibration/start_od_<time>.txt*).
- SUMO files to conduct measurements during simulations in the additional files (*calibration/Aachen_detectors.add.xml*)
- Observed traffic measurements, including traffic counts, average speed and density (all in *calibration/true_data_<time>.csv*).
- Traffic counts from 100 simulations (*calibration/t_test_counts100.csv*), which is used to derive the number of simulation replications to obtain a statistically significant result.

The following components are to be completed as part of this project. Each task is accompanied by specific expectations and hints to guide your work.

1. Evaluation of the minimum required number of simulation replications that give statistically significant outputs with at least 95% confidence level *(show calculations)*. You are asked to do the calculation based on link counts. Additionally, the desired tolerance of the counts for each link is set as 10%. Data from 100 simulation replications are available in *calibration/t_test_counts100.csv*. Please plot the result. (10 points)
2. Exploration in goodness-of-fit function, data measurement, value of information. (10 points)
   (a) Explain the superiority of Normalized Root Means Square Error (RMSN) compared to to other objective functions such as Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE), and discuss how GLS-based weighting schemes (i.e., Generalized Least Squares as a weighting/estimation method) differ from these error measures in large-scale OD demand calibration problems. (Hint: Consider the contribution difference of changes in traffic counts on busy links and low-flow links to the goodness-of-fit evaluation.) (5 points)
   (b) Evaluate the **starting RMSN** using different data types (link counts, speeds, and densities) and their combinations (with different weights) under different multiples of input OD. Namely, do a line search in input OD for different data combinations. To be specific, given the input OD as $\mathbf{x}$, try $0.2\mathbf{x}$, $0.4\mathbf{x}$, $0.6\mathbf{x}$, $0.8\mathbf{x}$, $\mathbf{x}$, $1.2\mathbf{x}$, $1.4\mathbf{x}$, $1.6\mathbf{x}$, $1.8\mathbf{x}$, $2\mathbf{x}$. Please plot the result. (Hint: Don't forget to normalize the weights to 1.) (5 points)
3. Calibrate the traffic model's demand for the morning peak hour (8 am - 9 am) using the given observed data (RMSN error should be at most 20% or less, better results will be encouraged). The calibration approach should be systematically documented covering different aspects e.g., traffic dissipation procedure, data processing, modeling assumptions (e.g., simulation parameters out of calibration scope), input space exploration, modifications to SPSA, calibration parameter search (e.g., using manual, random or automatic search) and results visualization. Students are encouraged to innovate as well as

adopt guidelines from scientific publications and also cite them. (Hint: Check the gradients to find appropriate SPSA parameters.) (30 points)

4. Bonus: Implement one of the three variants of SPSA: c-SPSA, w-SPSA, or PC-SPSA, and complete the same calibration task as in task 3. Compare and discuss the calibration performance of SPSA and its variants. (Bonus: 10 points)

3.2 Dynamic Applications (50 points in total)

In this task, you are asked to implement real-time traffic control measures to improve performance of a bus line in the network. In practice, you would start this task after fully calibrating your simulation model and run the number of replications derived in the model calibration part. However, this part of the project focuses on the dynamic application, and we want you to be able to work on it right away. Hence, for this part of the project, it is sufficient to run a single replication. Moreover, you should use the predefined demand files.

In this task, we analyze the impact of flexible bus stopping operations and traffic light prioritization for a section of bus line 3 in Aachen. The input files for this task are located in the *sumo_configs* directory and can be identified by the prefix *dyn_*. The corresponding configuration file is *dyn_config.sumocfg*. The bus schedule, vehicle types, passenger flow, and stops are already defined in the network and must not be modified. In the status-quo scenario, the traffic signals at the relevant junction operate with fixed-time control. The buses run at 5-minute headways and dwell for 30 seconds at each stop.

3.2.1 Dynamic Dwell Times

In reality, dwell times can fluctuate and may influence a bus's schedule. In this task, this behavior should be implmented in SUMO with the help of TraCI. The dwell time is thereby to be determined as a function of the number of passengers boarding and alighting at a stop. The following simplified model is provided for this purpose:

$$t_{\text{dwell}}(s) = 15 + 0.5 \cdot (n_{\text{board}}(s) + n_{\text{alight}}(s))$$

where $t_{\text{dwell}}(s)$ denotes the dwell time at stop s in seconds, and $n_{\text{board}}(s)$ and $n_{\text{alight}}(s)$ represent the number of passengers boarding and alighting at stop s, respectively.

1. Implement the dynamic dwell times in SUMO using the simplified model presented above and TraCI.

2. For each bus stop, determine the average dwell times over the entire time period $t = 0$ to $t = 3600$.

3. Based on the simulation results, discuss how the dynamic dwell times, compared to a constant dwell time of 30 seconds, affect the travel time and the reliability of the bus line.

3.2.2 Request Stops

In the status quo simulation, buses stop at every bus stop to allow passengers to board or alight. However, if no passenger wishes to get on or off, this results in unnecessary time and resource consumption. In bus systems, request stops are therefore often implemented, where the bus only stops if at least one passenger intends to board or alight. In practice, this is achieved through stop-request buttons and the driver's visual detection of waiting passengers. This behavior cannot be implemented directly in SUMO and requires the use of dynamic applications via TraCI.

1. Implement request stops using TraCI for the whole section of the bus line specified in the demand file, replacing fixed bus stops. Note that passengers appear randomly at stops over time and that the exact movements of passengers are not known to the driver before the bus approaches the next stop. The decision whether the bus should stop or not should be made 50–60 m before the bus reaches the stop.

2. Compare the impact of request stops on both passengers and bus operations with the status quo scenario by analyzing travel time and fuel consumption of the vehicles.

3.2.3 Public Transport Prioritization

Public transport prioritization is commonly used in practice to accelerate bus services. This approach has also be applied to the bus line in the present network.

1. Implement real-time control (with TraCI) for the signalized intersection of Krefelder Straße/Monheimsallee (Junction 60713745) along the bus route to accommodate bus prioritization. You can use strategies from Traffic Control lectures or develop your own. Either way, you should explain your approach in the presentation and the report.
    - The objectives are to avoid bus delays and to compensate green times for opposing traffic streams.
    - It is sufficient to have one registration process, i.e. it is not necessary to have pre-registration and registration.
    - You can assume an uncongested traffic state, i.e. a maximum green time for the extension does not have to be considered.
2. Discuss the effectiveness of the implemented measure by comparing the average travel time for both passenger cars and buses (split by traffic streams along the corridor and crossing streams).
3. To support your analysis, plot the time-space diagram along the corresponding corridor for each bus individually. Please include the green phases of the signals and other car trajectories along the corridor in the plot. A selection of these plots showing examples for the application of the different control strategies should be included in the report.

4 Report

The report should have the following chapters:

1. Introduction: you should briefly put this work into context; **use your own words** to describe the necessity of calibrating microscopic traffic simulations and motivations for both dynamic applications.
2. Calibration: you should present your methods and model calibration results here. It should contain an introductory analysis, error and objective function plots, and comparisons of original and calibrated data as well as target and calibrated data. You can also show other characteristics if you deem them important.
3. Dynamic Application: you should present the used methodology, including a brief description of the TraCI implementation details, as well as the results of the applied measures.

Here are some general comments on report writing:

- Consider a student in the field before this course as a reader to distinguish between basic knowledge, which you can simply mention, and methods you should explain in more detail
- Figures should have readable axes, legends, and captions that put them into context
- Figures should clarify your points; explain and interpret these points in the respective text
- Write clearly and precisely, consider using a language tool (such as Grammerly) to check the grammar and spelling of your report

5 Submission Procedure

Please collect your report and presentation (pdf format), as well as your code and input files in a zip file with the name 'MTS_2526_LASTNAME_FIRSTNAME.zip'. Result files (especially large picture files) should not be included. The zip file has to be submitted on the Moodle platform in the *Special Topics on Model Calibration* course. Please submit your work on time. **The online submission form will close with the deadline!**